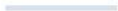


Reliable scikit-learn Pipelines

Encode domain knowledge without leaking
information across model boundaries.



Learning outcomes

- 01  Fit preprocessing only on training data.
- 02  Compose column transforms and estimators.
- 03  Persist and reload a complete inference artifact.

The concept system

01

fit and transform

02

ColumnTransformer

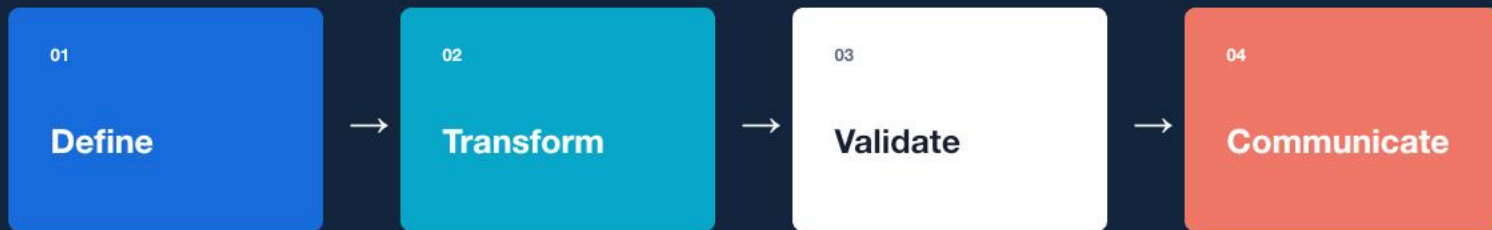
03

Pipeline

04

serialization

A repeatable reasoning loop



Build a mixed-type preprocessing and model pipeline.

Guided lab

LAB BRIEF

Build a mixed-type preprocessing and model pipeline.

workbook / tests / evidence

01

Frame

Define inputs, outputs, and success criteria.

02

Build

Implement the smallest correct workflow.

03

Verify

Test normal, boundary, and failure cases.

04

Explain

Record the evidence and remaining limits.

Portfolio deliverable

A fitted pipeline, inference test, and feature contract.

QUALITY GATE

-  **Reproducible**
-  **Tested**
-  **Decision-relevant**

WEEKLY ASSESSMENT

Leakage-Safe Feature Pipeline

Boundary correctness 35%,
rationale 25%, reproducibility
20%, tests 10%, explanation
10%.

What should remain after the lesson?

CONCEPT

fit and transform + ColumnTransformer

PRACTICE

Build a mixed-type preprocessing and model pipeline.

EVIDENCE

A fitted pipeline, inference test, and feature contract.
